KL UNIVERSITY • BACHUPALLY CAMPUS, HYDERABAD

Department of Computer Science & Engineering

REAL-TIME SIGN LANGUAGE INTERPRETER

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Introduction & Domain Background

SignVision is a machine learning-based system designed to recognize hand gestures representing sign language alphabets and convert them into readable text. The system uses a labeled image dataset to train models that can identify different hand signs. It combines Machine Learning, Computer Vision, and Image Processing to perform gesture recognition. The project aims to explore how intelligent systems can help reduce communication barriers and support accessibility for people who use sign language.

Problem Statement & Motivation

Communication between sign language users and people who do not understand sign language can be challenging, especially when an interpreter is unavailable. This may create difficulties in everyday situations such as education, public services, and social interactions. SignVision aims to address this problem by recognizing hand gestures and displaying their corresponding alphabets as text. The motivation behind this project is to develop a simple and accessible application that demonstrates how machine learning can support easier communication.

Literature Survey

Several research studies have explored machine learning and deep learning techniques for sign language and hand gesture recognition. Traditional algorithms such as Logistic Regression and Random Forest can be applied to classification tasks using suitable image features. Convolutional Neural Networks (CNNs) are particularly useful for image classification because they can learn visual patterns directly from images. Research also highlights the importance of preprocessing, data augmentation, and proper model evaluation in developing reliable recognition systems. These approaches form the foundation of the SignVision project.

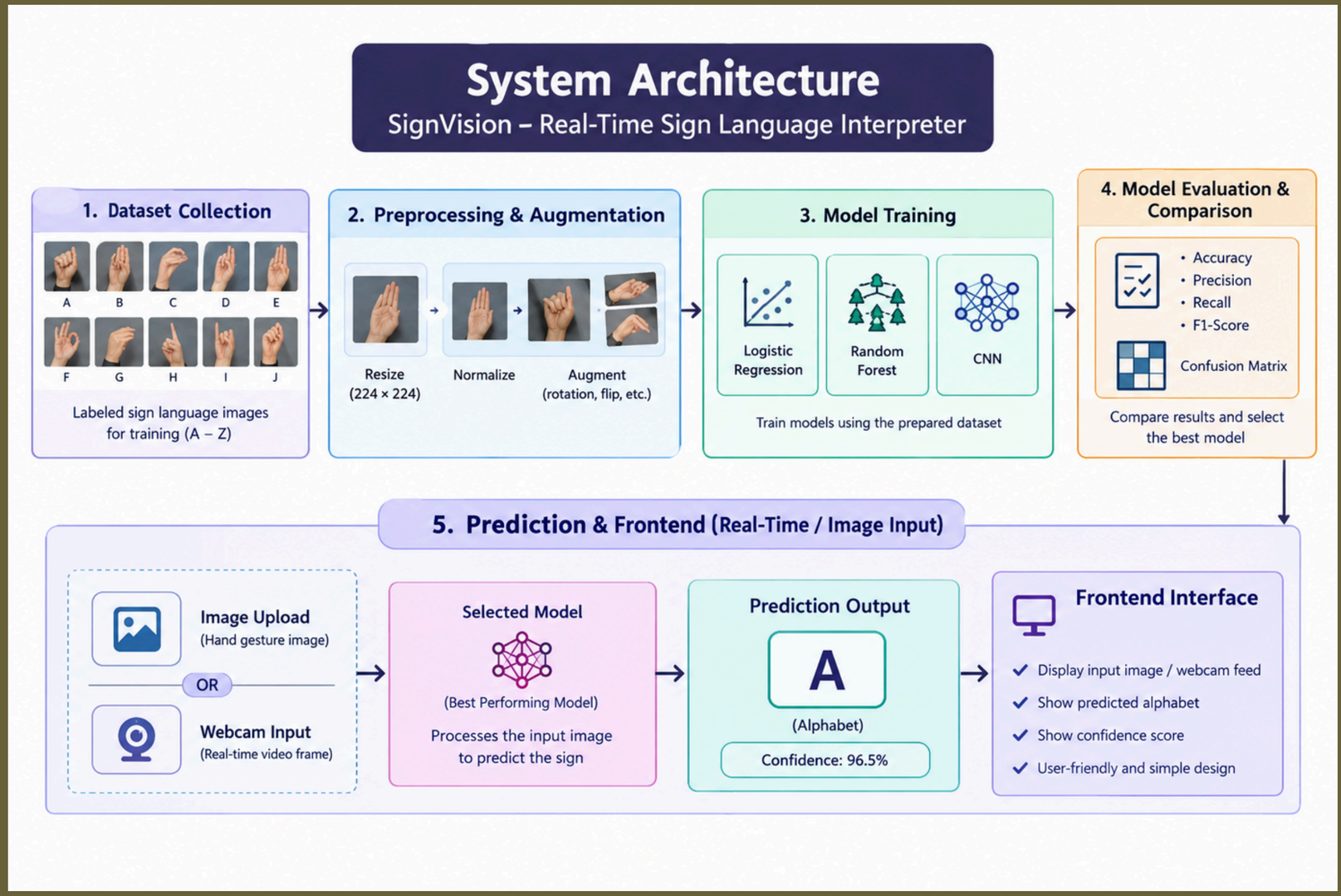
Research Gap Identified

Although different techniques are available for sign language recognition, their performance can vary depending on the dataset, lighting conditions, image quality, and hand position. Some approaches focus on a single classification model, making it difficult to understand how alternative methods perform under similar conditions. There is a need to compare multiple models using the same dataset and consistent evaluation metrics. SignVision addresses this aspect by comparing Logistic Regression, Random Forest, and CNN models and exploring their use in image-based and webcam-based sign recognition.

Objectives of the Project

- To develop a system that recognizes sign language alphabets from hand gesture images.
- To preprocess and augment the dataset to prepare images for effective model training.
- To implement and compare Logistic Regression, Random Forest, and CNN models.
- To evaluate model performance using accuracy, precision, recall, F1-score, and a confusion matrix.
- To develop a simple frontend for image upload, webcam input, and displaying predicted signs.

Proposed System Architecture



Methodology & Modules

- Dataset Module: Loads labeled hand gesture images and their corresponding alphabet labels.
- Preprocessing Module: Resizes and prepares images, with augmentation applied where appropriate.
- Model Training Module: Trains Logistic Regression, Random Forest, and CNN for sign classification.
- Evaluation Module: Measures and compares model performance using standard classification metrics.
- Prediction Module: Processes uploaded images or webcam frames and predicts the corresponding sign.
- Frontend Module: Provides an interface to upload images, access webcam input, and display predictions.

Tools and Technologies

- Programming Language: Python for data processing, model training, and prediction.
- Machine Learning: Scikit-learn for Logistic Regression and Random Forest; TensorFlow/Keras for CNN.
- Image Processing: OpenCV and NumPy for image handling and preprocessing, i used in the implementation.
- Data Analysis: Pandas and Matplotlib for dataset handling and performance visualization.
- Frontend Development: HTML, CSS, and JavaScript for creating the user interface.
- Development Environment: VS Code or Jupyter Notebook for coding, training, and testing.

Work Completed / Demo

- The project problem, objectives, and overall system workflow have been identified.
- Logistic Regression, Random Forest, and CNN have been selected as the approaches for model comparison.
- The dataset preparation, preprocessing, training, and evaluation workflow has been planned.
- The frontend is planned to support image upload, webcam input, and predicted sign display.
- The final demo will showcase the trained model's predictions and evaluation results once implementation is complete.

Timeline & Review Plan

- Phase 1: Identify the problem, define objectives, and conduct the literature survey.
- Phase 2: Collect the dataset and perform image preprocessing and augmentation.
- Phase 3: Train Logistic Regression, Random Forest, and CNN models.
- Phase 4: Evaluate the models, compare results, and select the best-performing approach.
- Phase 5: Develop the frontend and integrate the selected trained model.
- Phase 6: Test the complete application, prepare the final demo, and document the results.

Expected Outcome & Conclusion

- The project is expected to recognize sign language alphabets from input hand gesture images.
- The comparison will help identify which of the selected models performs best on the chosen dataset.
- The frontend will aim to provide predictions from uploaded images and webcam input.
- Evaluation metrics will help measure the reliability and classification performance of the models.
- SignVision demonstrates the potential of machine learning to support accessible communication.

References

- Scikit-learn Documentation. https://scikit-learn.org/
- TensorFlow Documentation. https://www.tensorflow.org/
- Keras Documentation. https://keras.io/
- OpenCV Documentation. https://docs.opencv.org/
- Research papers on sign language recognition, image classification, and CNN-based gesture recognition from IEEE Xplore and Google Scholar.

THANK YOU